
ENGINEERING SPECIFICATIONS

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**COST EFFECTIVE
CHILLER PLANT MANAGER**

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1. Introduction

Aspiration for energy conservation and necessity to reduce manual intervention in plant operation to minimum level, demands for Plant Manager (PM). At the same time the recent developments in programmable controllers and control equipment resulting in low cost and high reliability made it practically feasible.

Kirloskar Chillers Pvt. Ltd. has developed the Plant Manager, flexible enough to accommodate various combinations of plant equipment & there operating sequences and smart enough to stage/operate them to optimize for power and performance.

2. CPM Details

2.1. Hardware Description

1. Carel make PLC control system
2. Eldon/Rittal/Other make wall mounted MS enclosure with required PLC & modules + required switchgear mounted and completely wired up till the terminals.
3. PVR make 15" HMI display of resolution 1920x1080,
4. Renu make field IO's.

2.2. Software Description

1. Carel 1Tool for PLC programming
2. FSstudio for HMI designing
3. Delta Dia-View SCADA.(at separate/additional cost)

4. Plant Description

Chiller Plant Manager (CPM) controls/operates following equipment in plant,

- a. Water / Air cooled chillers
- b. Heat Pump
- c. Chilled Water / Evaporator pump
- d. Chilled Water / Evaporator valve
- e. Condenser pump
- f. Condenser valve
- g. Secondary Chilled Water Pump
- h. Cooling tower fan
- i. Cooling tower Inlet/Outlet valve

Apart from above equipment following monitoring / measuring field instruments are also considered

- 1. Chilled water flow
- 2. Chiller Header Temp
- 3. Condenser Header Temp
- 4. AHU Pressure
- 5. Differential Pressure
- 6. Bypass Valve
- 7. Pump VFD

Note - Above all instruments may or may not present in every project, it depends on specifications of individual project.

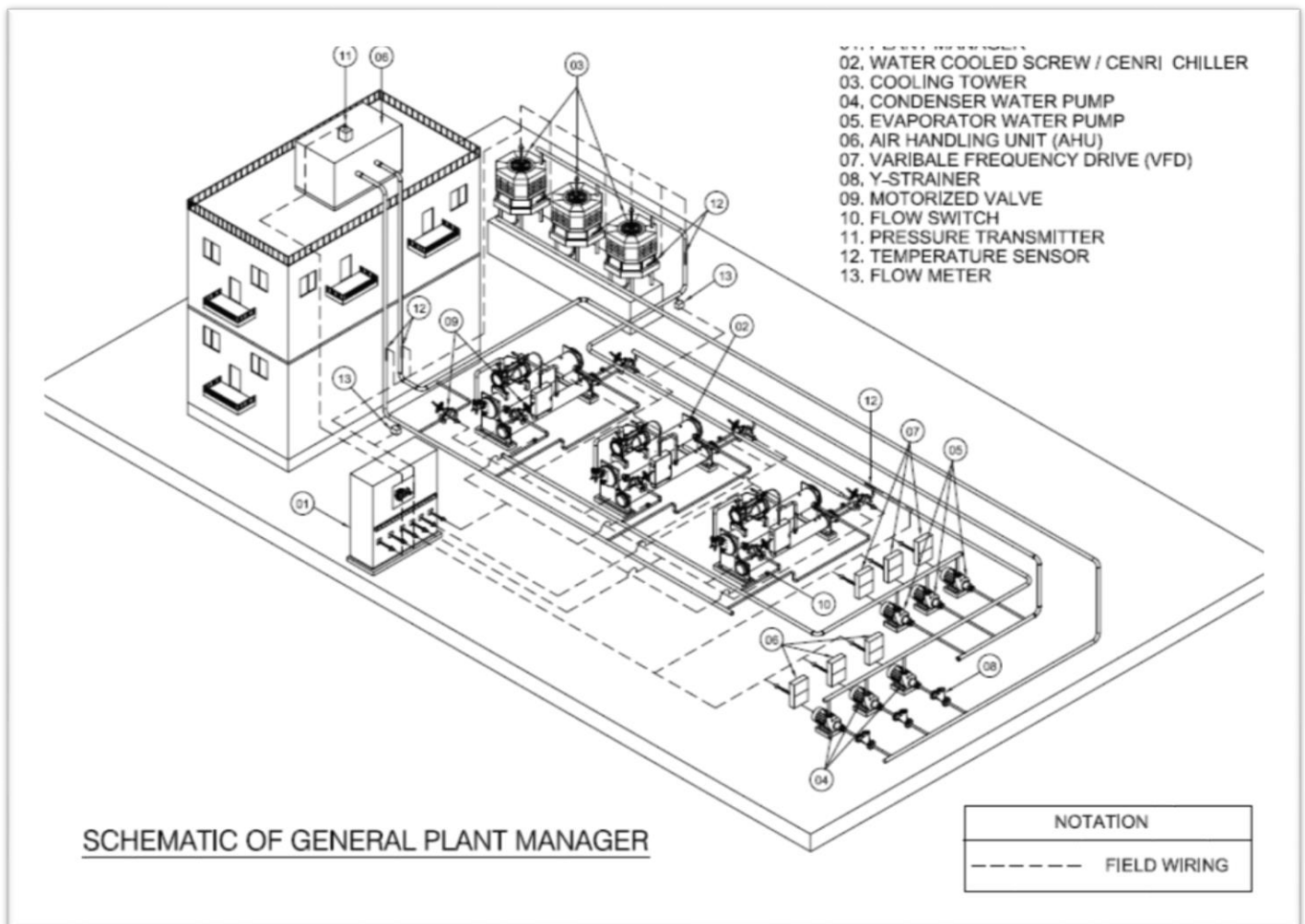


Fig.3 Sample Schematic of Plant Manager

5. Functionality

A. General

Chillers will be operated in local mode & remote mode; local/remote mode selection shall be done from chiller controller. Remote mode operation shall be initiated from Plant-Manager (PM) by start/stop command to chiller control panel. During Local Mode Plant-Manager (PM) intervention shall be bypassed.

Chillers will be sorted based on run-hours and healthy feedback of them along with all allied items (pumps, valves etc.). Trip feedback of any one item for corresponding chiller, makes it unavailable. First number will be assigned to chiller with least run-hours and similarly to other available chillers. Start-up of chiller in algorithm will also mean starting corresponding pumps & cooling tower.

Chilled water isolation valve & Chiller are together considered as one group since failure of any one will make other unavailable, whereas two such groups are independent of each other for healthiness. Same is the case with evaporator pumps since they are connected with each other via common header system, so there can be any pump associated with any group of chiller & valve. Alarm condition (Trip, fail to Start/Open and Fail to Stop/Close) of any element in group is considered as alarm for that group and under such conditions that particular group is immediately replaced by next available alternate group.

B. Staging of Chillers

Single set point and a dead band are set to decide loading and unloading profile with respect to supply water temperature value. Deadband is equally distributed across setpoint so that loading zone is for supply water temperature $> \text{Setpoint} + \text{Deadband}/2$ while unloading zone is for supply water temperature $< \text{Setpoint} - \text{Deadband}/2$. While temperature is in deadband, no loading or unloading is done.

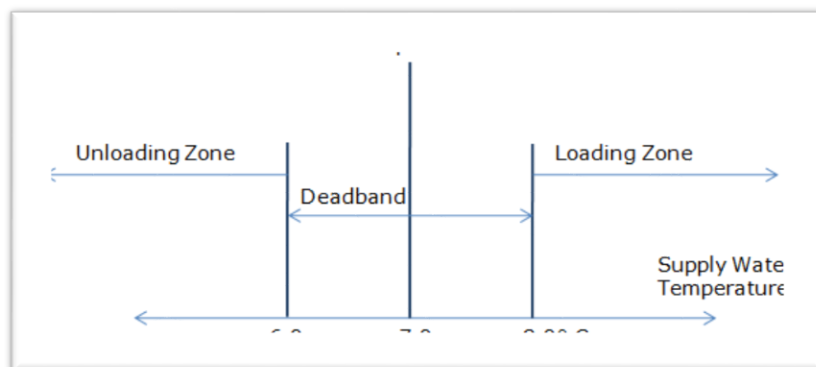


Fig.4 Chiller Setpoint

1. Loading sequence for Chillers

Plant manager needs start/Auto command to start the operation. Once it is started, status of all the equipment is evaluated and correct sequence is set. In case of chillers, sequence is set in such a way that the chiller with lowest run-hours is set at first to start and others in ascending order ahead. After starting up, one evaporator pump along with evaporator valves of first chillers are turned on irrespective of demand. If temperature is in loading zone after startup, demand is generated for chillers one by one using PID control logic. Demand for next chiller is generated only when already running chillers cross set limit of loading or run for preset time and delay time between demand steps is elapsed. Staging of pumps will directly be governed by staging of chillers.

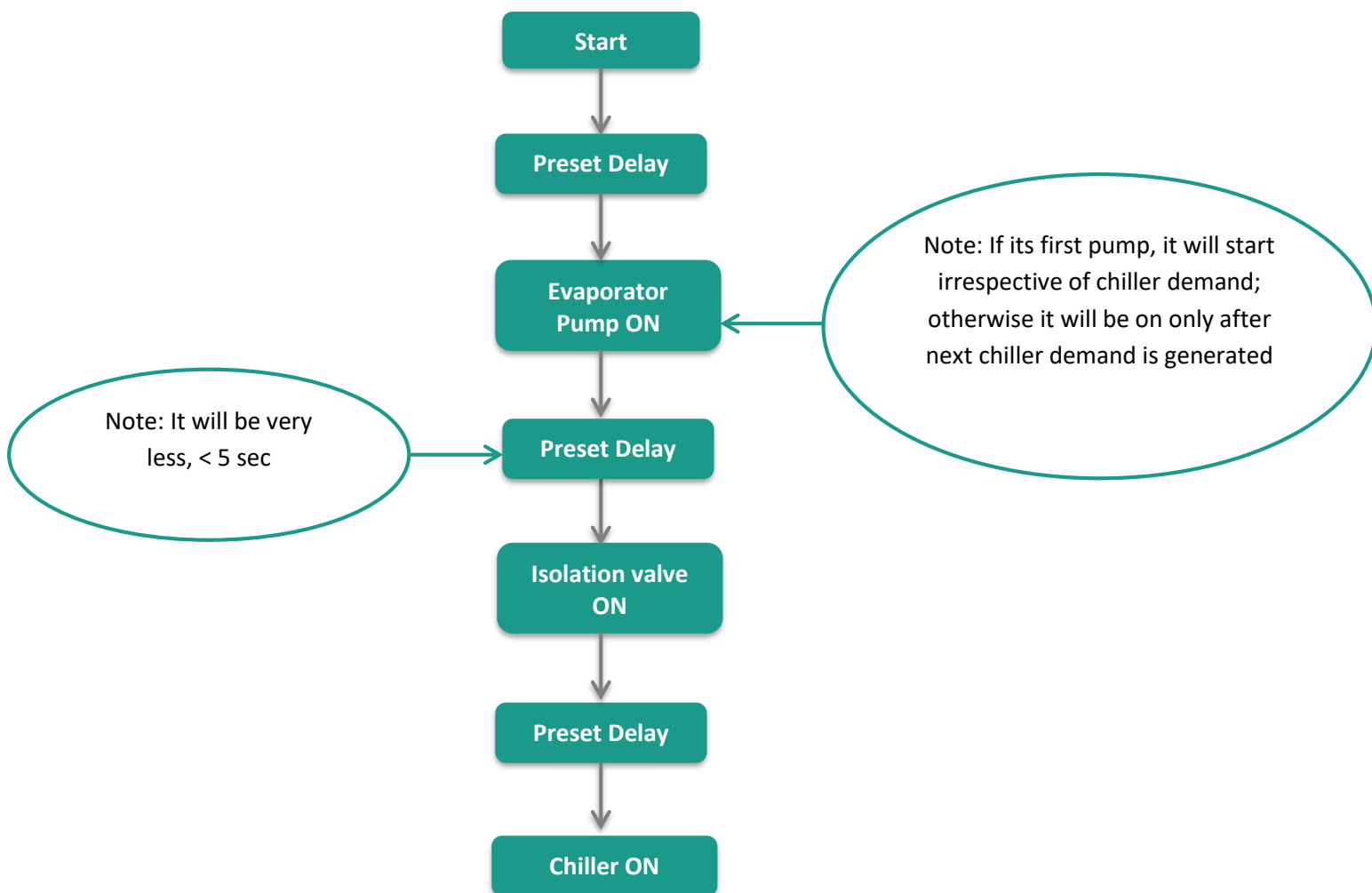


Fig.5 Loading Sequence

2. Unloading sequence for chillers

When supply water temperature is in unloading zone chiller units are turned off one by one in such a way that the chiller with highest run-hours is turned off first and others in descending order ahead. If temperature is in Unloading zone, demand is removed for chillers one by one using PID control logic. Selected chiller is turned off only when running chillers are unloaded to preset level. After turning off one unit there is delay time which must be elapsed before checking condition for turning off next chiller. Even if chiller is OFF corresponding valve and pump will be ON for preset delay.

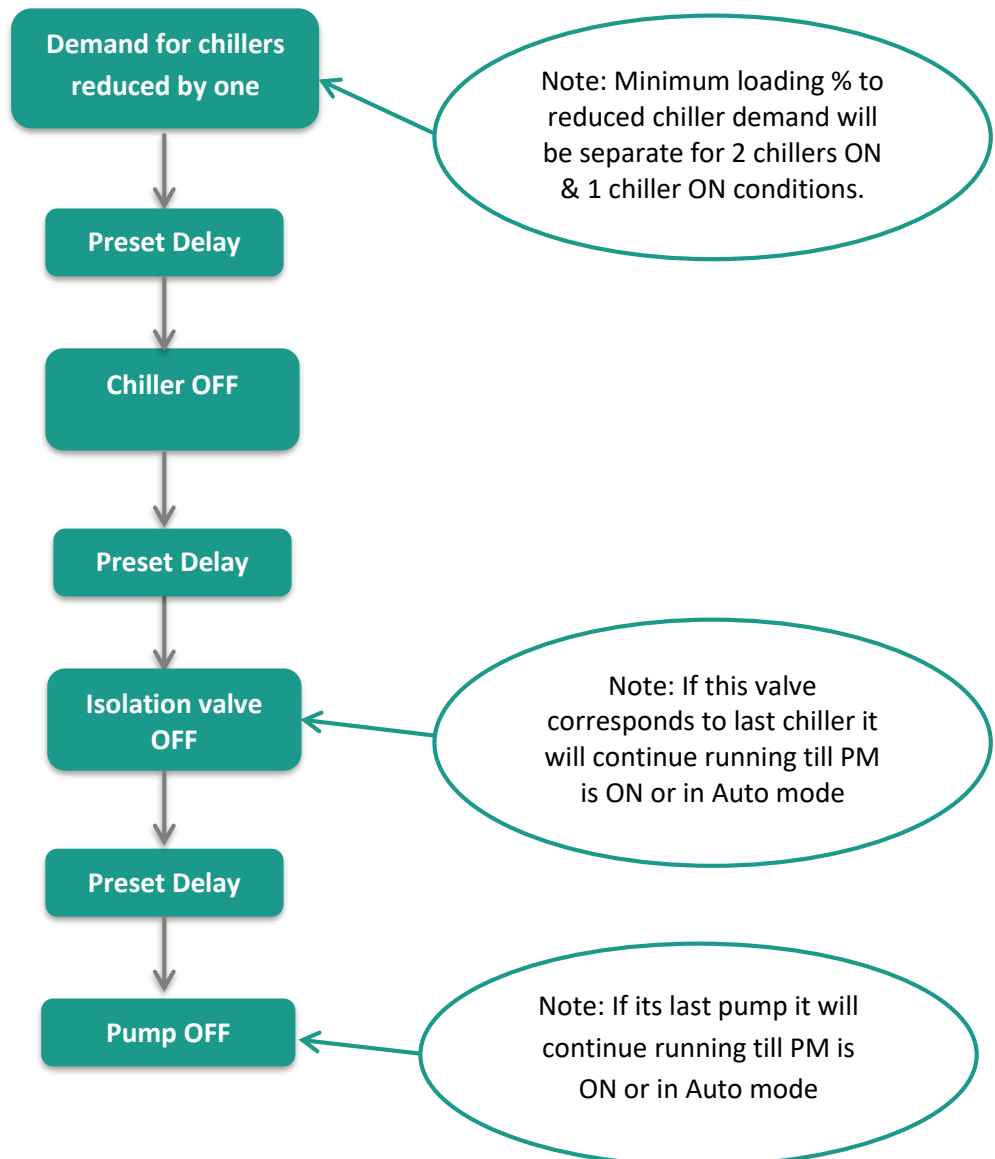


Fig.6 Unloading Sequence

C. Modulation of Pumps

Secondary loop design is implemented between chiller and actual load. The plant contains two different water loops; Primary – circulating through chiller & primary pumps and Secondary- circulating through load & secondary pumps.

Chilled water Pump Logic for Variable Flow system

1. Given 3 no. of chilled water pumps supplying to common header line. Single VFD to be shared by pumps for speed control.
2. PID controller will control ON/OFF and speed for all CW pumps so as to maintain Differential pressure across AHU. Set point for controller can be set from HMI.
3. At any given point of time maximum 2 pumps will be turned on to meet the set pressure. In this sequence Pump with VFD starts first with Minimum speed and goes to maximum as demand rises. When it reaches maximum speed second pump is turned ON and first pump speed comes to minimum and starts rising again with demand.
4. Pump getting associated with VFD is rotated every week.
5. As Diff pressure decreases below set value, controller will increase Speed/No of pumps to maintain the pressure and vice versa.
6. While Plant Manager is ON minimum one pump will continue to run with min speed even if pressure demand is met.
7. Minimum/Maximum speed limit for pumps can be set from HMI
8. Distribution of Digital and analog output over PID output can be represented as below

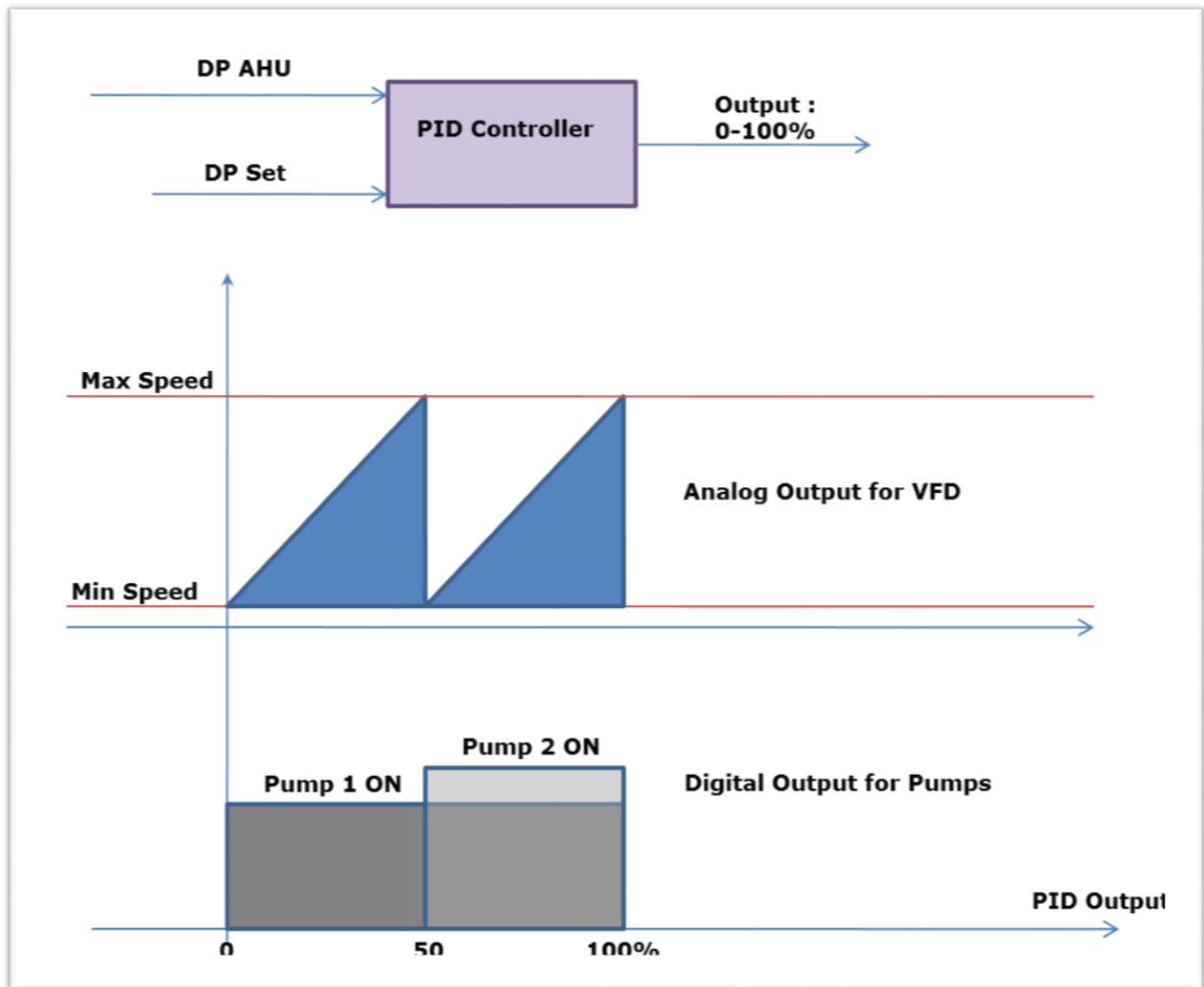


Fig.7 Chilled Water Pump working

6. Operation

Chiller Plant Manager System controls the chiller system and its units which comprises of Evaporator pumps and valves, Condenser pumps and Valves, and Cooling Tower fan and valves.

6.1. Power on Sequence

- MCB's in panel: -
MCB 1 & 3 is used for 230VAC.
MCB 2 24VDC Panel Supply.
- Sequence: -
1st turn ON MCB 1 & 3, then turn ON MCB 2. For Turning OFF follow same procedure in reverse order.

6.2. Operating Modes

Chiller Plant Manager operates in 2 modes, i.e. AUTO/MANUAL. Selection is provided on HMI Panel. For AUTO mode PLC will take care of control process for Chillers. In MANUAL mode individual START/STOP push buttons for individual equipment's are provided on HMI Panel for control.

HMI is used for monitoring & control for process.

HMI screens are as follows: -

- 1) Startup Screen
- 2) Overview
- 3) Evaporator Pumps
- 4) Cooling Tower
- 5) Condenser Pumps
- 6) Chillers
- 7) Settings
- 8) Alarms
- 9) Alarm History
- 10) Modbus Data

Online alarms are provided on top of every screen with reset button on each screen.

6.2.1. Startup Screen

After giving power supply (24VDC) to HMI, HMI will start with the Startup screen as a home screen.



This screen is actual startup screen for the HMI panel. Whenever the Panel is powered ON or the HMI is Powered ON this Screen will be displayed. It has a Startup Time which is activated when PLC is Powered ON. This is a non-settable 60 seconds delay after which PLC starts execution of its program, i.e. it gives its command to equipment's connected to the system.

Startup screen appears on the HMI only for once, if any other screen button is pressed then this screen can be viewed only after power cycle of the HMI. HMI screen Navigation tabs are provided for other screens. Respective screen is shown on pressing the respective tabs.

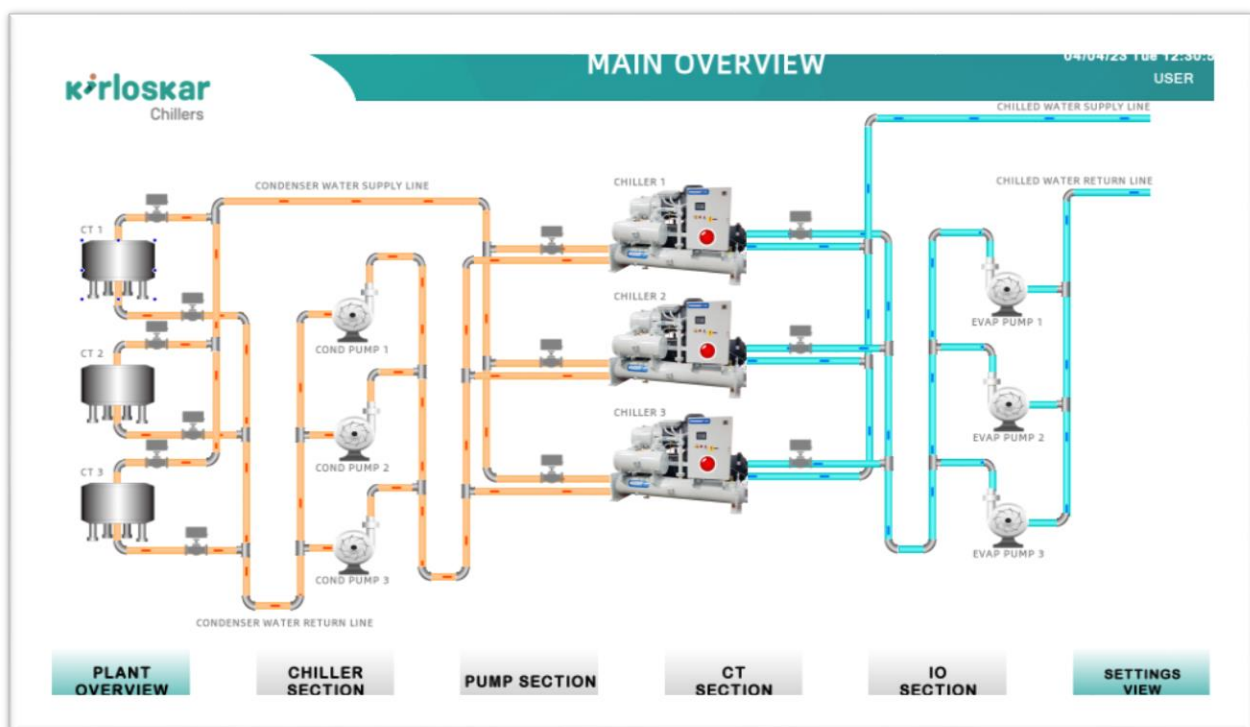
6.2.2. Overview

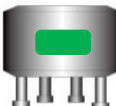
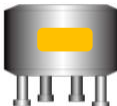
On pressing the OVERVIEW tab, the following screen will appear on HMI Panel. This screen provides an overview of the system, there are two navigation buttons provided on this screen for navigation of screens within the OVERVIEW screen.

List of screens in OVERVIEW screen: -

a) Chiller Overview

The following screen shows you status of chiller evaporator equipment, Supply header temp and Return header temp. There 3 buttons provided to view Modbus data of all 3 chillers respectively. Below the screen diagrammatic representation shows the status and their respective color assignments.



PUMP			
VALVE			
CHILLER			
CT FAN			

i. Analog Input

MAIN PANEL CONTROLLER ANALOG INPUT		
U-1	Flow Meter Inlet	123.4
U-2	Flow Meter Outlet	123.4
U-3	SPARE	123.4
U-4	CHILLED WATER HEADER SUPPLY TEMP	123.4 °C
U-5	CHILLED WATER HEADER RETURN TEMP	123.4 °C
U-6	CONDENSER HEADER RETURN TEMP	123.4 °C
U-7	CONDENSER HEADER RETURN TEMP	123.4 °C

EX- 2 ANALOG INPUT CARD		
AI-0	Secondary Pump Absolute Pressure Transmitter 2	123.4 bar
AI-1	Secondary Pump Absolute Pressure Transmitter 3	123.4 bar
AI-2	Secondary Differential pressure sensor across Pump 1	123.4 bar
AI-3	Secondary Differential pressure sensor across Pump 2	123.4 bar
AI-4	Secondary Differential pressure sensor across Pump 3	123.4 bar
AI-5	Condenser Pump Absolute Pressure Transmitter 1	123.4 bar
AI-6	Condenser Pump Absolute Pressure Transmitter 2	123.4 bar
AI-7	Condenser Pump Absolute Pressure Transmitter 3	123.4 bar

ANALOG
INPUT'S

EX- 1 ANALOG INPUT CARD		
AI-0	Common header CDW supply pressure	123.4 bar
AI-1	Common header CDW return pressure	123.4 bar
AI-2	Common header CHW supply pressure	123.4 bar
AI-3	Common header CHW return pressure	123.4 bar
AI-4	Primary Pump Absolute Pressure Transmitter 1	123.4 bar
AI-5	Primary Pump Absolute Pressure Transmitter 2	123.4 bar
AI-6	Primary Pump Absolute Pressure Transmitter 3	123.4 bar
AI-7	Secondary Pump Absolute Pressure Transmitter 1	123.4 bar

DIGITAL
INPUT'S

PLANT
OVERVIEW

CHILLER
SECTION

PUMP SECTION

CT
SECTION

IO
SECTION

DIGITAL
OUTPUT'S

EX- 1 RIO PANEL AI CARD			EX- 2 RIO PANEL AI CARD		
AI-0	AMBIENT TEMPERATURE	123.4 °C	AI-0	SPARE	123.4
AI-1	AMBIENT HUMADITY	123.4 %	AI-1	SPARE	123.4
AI-2	SUMP Outlet Temperature 1	123.4 °C	AI-2	SPARE	123.4
AI-3	SUMP Outlet Temperature 2	123.4 °C	AI-3	SPARE	123.4
AI-4	SUMP Outlet Temperature 3	123.4 °C	AI-4	SPARE	123.4
AI-5	SPARE	123.4	AI-5	SPARE	123.4
AI-6	SPARE	123.4	AI-6	SPARE	123.4
AI-7	SPARE	123.4	AI-7	SPARE	123.4

ANALOG INPUT'S

DIGITAL INPUT'S

DIGITAL OUTPUT'S

← **PLANT OVERVIEW** **CHILLER SECTION** **PUMP SECTION** **CT SECTION** **IO SECTION** →

ii. Digital Input

MAIN PANEL CONTROLLER DIGITAL INPUTS		
DI-1	CHILLER 1 REMOTE/LOCAL STATUS	ON
DI-2	CHILLER 1 RUN STATUS	ON
DI-3	CHILLER 1 TRIP STATUS	ON
DI-4	CHILLER 2 REMOTE/LOCAL STATUS	ON
DI-5	CHILLER 2 RUN STATUS	ON
DI-6	CHILLER 2 TRIP STATUS	ON
DI-7	CHILLER 3 REMOTE/LOCAL STATUS	ON
DI-8	CHILLER 3 RUN STATUS	ON
DI-9	CHILLER 3 TRIP STATUS	ON
DI-10	EVAPORATOR OUTLET VALVE 1 OPEN STATUS	ON
DI-11	EVAPORATOR OUTLET VALVE 1 CLOSE STATUS	ON
DI-12	EVAPORATOR OUTLET VALVE 2 OPEN STATUS	ON
DI-13	EVAPORATOR OUTLET VALVE 2 CLOSE STATUS	ON
DI-14	EVAPORATOR OUTLET VALVE 3 OPEN STATUS	ON

MAIN PANEL DIGITAL INPUT CARD - 1		
DI-1	EVAPORATOR OUTLET VALVE 3 CLOSE STATUS	ON
DI-2	CONDENSER OUTLET VALVE 1 OPEN STATUS	ON
DI-3	CONDENSER OUTLET VALVE 1 CLOSE STATUS	ON
DI-4	CONDENSER OUTLET VALVE 2 OPEN STATUS	ON
DI-5	CONDENSER OUTLET VALVE 2 CLOSE STATUS	ON
DI-6	CONDENSER OUTLET VALVE 3 OPEN STATUS	ON
DI-7	CONDENSER OUTLET VALVE 3 CLOSE STATUS	ON
DI-8	PRIMARY PUMP RUN STATUS 1	ON

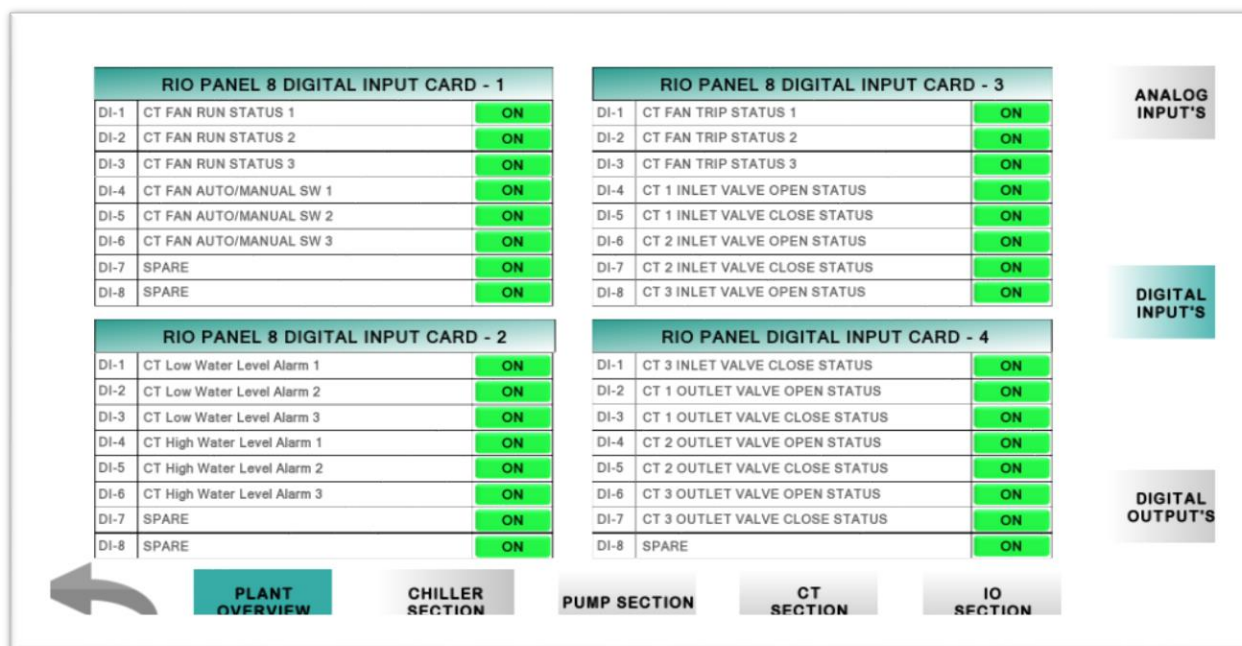
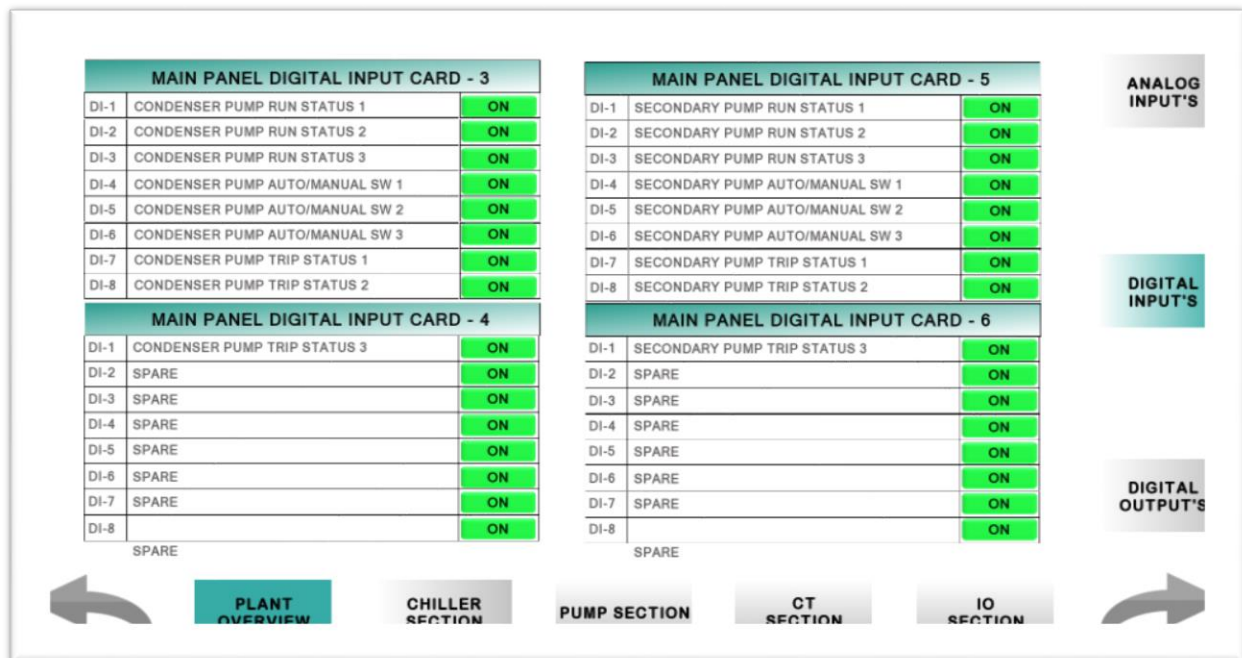
MAIN PANEL DIGITAL INPUT CARD - 2		
DI-1	PRIMARY PUMP RUN STATUS 2	ON
DI-2	PRIMARY PUMP RUN STATUS 3	ON
DI-3	PRIMARY PUMP AUTO/MANUAL SW 1	ON
DI-4	PRIMARY PUMP AUTO/MANUAL SW 2	ON
DI-5	PRIMARY PUMP AUTO/MANUAL SW 3	ON
DI-6	PRIMARY PUMP TRIP STATUS 1	ON
DI-7	PRIMARY PUMP TRIP STATUS 2	ON
DI-8	PRIMARY PUMP TRIP STATUS 3	ON

ANALOG INPUT'S

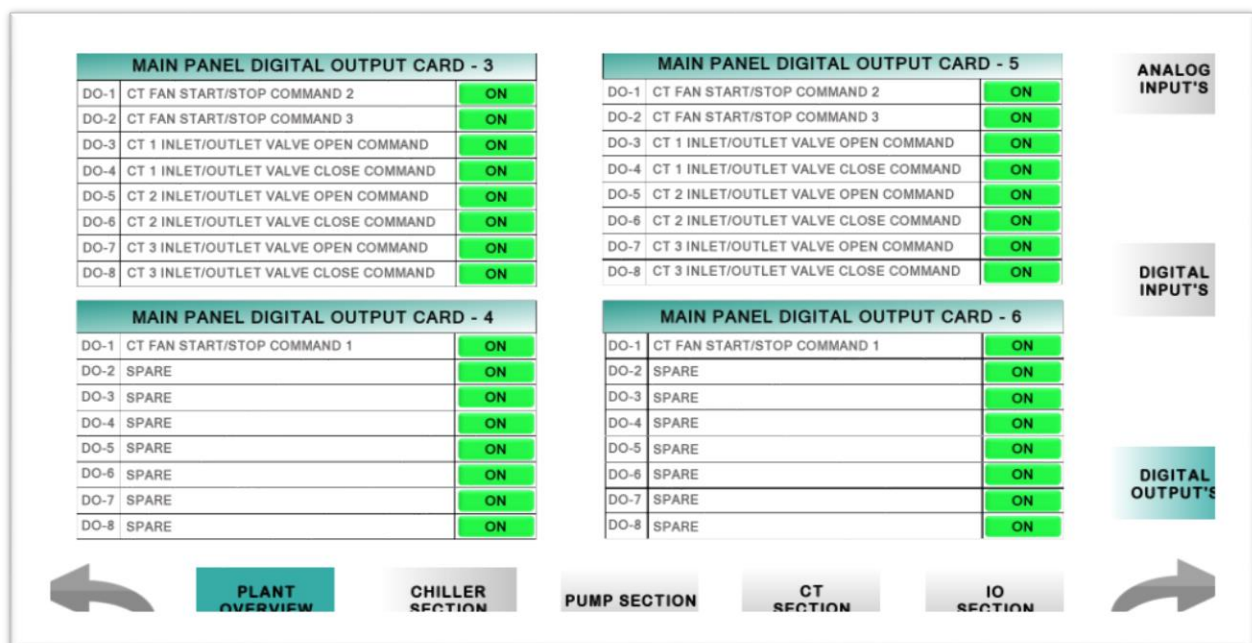
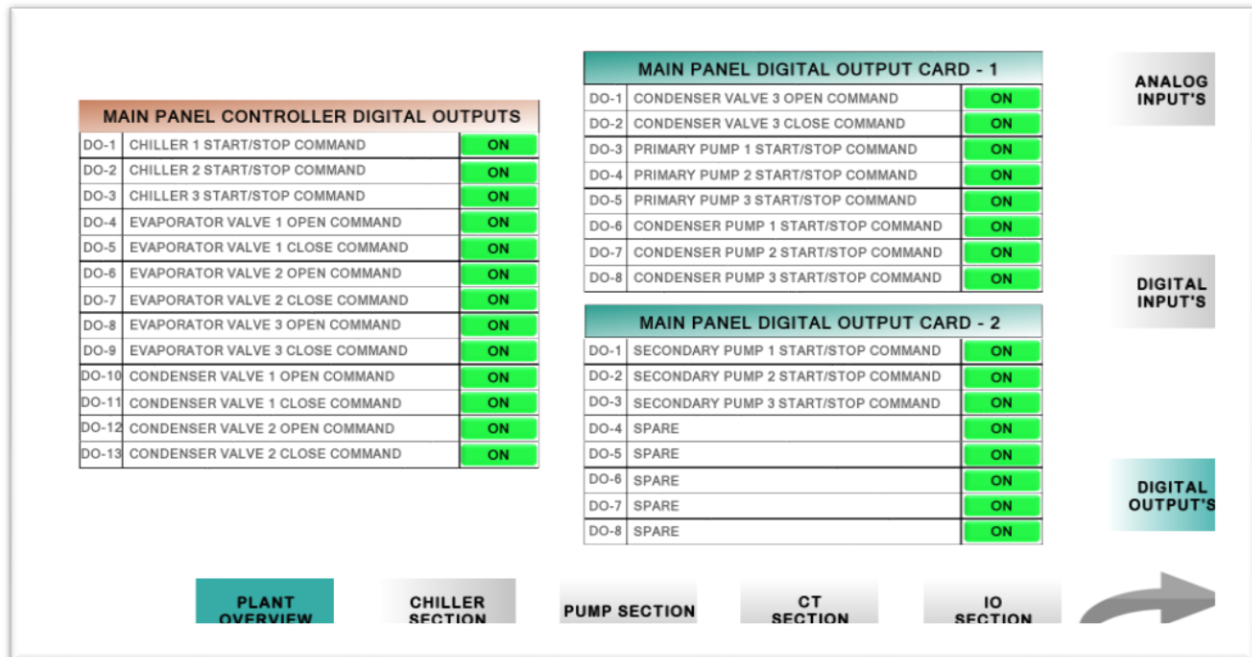
DIGITAL INPUT'S

DIGITAL OUTPUT'S

← **PLANT OVERVIEW** **CHILLER SECTION** **PUMP SECTION** **CT SECTION** **IO SECTION** →



iii. Digital Outputs



RIO PANEL DIGITAL OUTPUT CARD - 1			ANALOG INPUT'S
DO-1	CT FAN START/STOP COMMAND 2	ON	
DO-2	CT FAN START/STOP COMMAND 3	ON	
DO-3	CT 1 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-4	CT 1 INLET/OUTLET VALVE CLOSE COMMAND	ON	
DO-5	CT 2 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-6	CT 2 INLET/OUTLET VALVE CLOSE COMMAND	ON	
DO-7	CT 3 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-8	CT 3 INLET/OUTLET VALVE CLOSE COMMAND	ON	
RIO PANEL DIGITAL OUTPUT CARD - 2			DIGITAL INPUT'S
DO-1	CT FAN START/STOP COMMAND 1	ON	
DO-2	SPARE	ON	
DO-3	SPARE	ON	
DO-4	SPARE	ON	
DO-5	SPARE	ON	
DO-6	SPARE	ON	
DO-7	SPARE	ON	
DO-8	SPARE	ON	
RIO PANEL DIGITAL OUTPUT CARD - 3			DIGITAL OUTPUT'S
DO-1	CT FAN START/STOP COMMAND 2	ON	
DO-2	CT FAN START/STOP COMMAND 3	ON	
DO-3	CT 1 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-4	CT 1 INLET/OUTLET VALVE CLOSE COMMAND	ON	
DO-5	CT 2 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-6	CT 2 INLET/OUTLET VALVE CLOSE COMMAND	ON	
DO-7	CT 3 INLET/OUTLET VALVE OPEN COMMAND	ON	
DO-8	CT 3 INLET/OUTLET VALVE CLOSE COMMAND	ON	
RIO PANEL DIGITAL OUTPUT CARD - 4			
DO-1	CT FAN START/STOP COMMAND 1	ON	
DO-2	SPARE	ON	
DO-3	SPARE	ON	
DO-4	SPARE	ON	
DO-5	SPARE	ON	
DO-6	SPARE	ON	
DO-7	SPARE	ON	
DO-8	SPARE	ON	

PLANT OVERVIEW
CHILLER SECTION
PUMP SECTION
CT SECTION
IO SECTION

6.2.3. Evaporator Pumps

Evaporator screen shows the conditions of the evaporator pump related to Pump 1, Pump 2 and Pump3. It indicates the fault which is generated during the operation. When Mode is LOCAL no operation of the respective pump is possible in Auto or Manual mode. Navigation buttons are to be used to navigate through

PUMP OPERATION					
EVAPORATOR PUMP'S					
	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
EVAP PUMP 1	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
EVAP PUMP 2	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
EVAP PUMP 3	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
CONDENSER PUMP'S					
	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
COND PUMP 1	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
COND PUMP 2	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
COND PUMP 3	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
SECONDARY PUMP'S					
	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
SEC PUMP 1	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
SEC PUMP 2	UNHEALTHY	LOCAL	STOPPED	MANUAL	START
SEC PUMP 3	UNHEALTHY	LOCAL	STOPPED	MANUAL	START

PLANT OVERVIEW
CHILLER SECTION
PUMP SECTION
CT SECTION
IO SECTION

Evaporator Pump Overview

the screen for Pump 2 and Pump 3.

i. Evaporator Pump 1

To change the operation of the pump from AUTO to MANUAL click on MANUAL button, a LOG ON window will appear on the screen. If you have Authorization for this level of access, then enter your ID and Password.

It is recommended to be very careful while switching from Auto to manual when device is ON.

If the device is ON, then after switching from AUTO to MANUAL it will remain ON due to BUMPLESS TRANSFER. No two device of same type should be transferred from AUTO to MANUAL at a time.

Condition 1 - When 1 or 2 chillers are ON: -

Switching the ON device from Auto to Manual will turn the next OFF device in sequence ON, and the chiller demand will remain same.

Condition 2 - When 3 chillers are ON: -

Switching the ON device from Auto to Manual, a settable delay timer will start counting before which all the other device of one respective sequence should be transferred from Auto to Manual by the operator, failing to do so will cause the tripping of chiller. Chiller demand is also decreased.

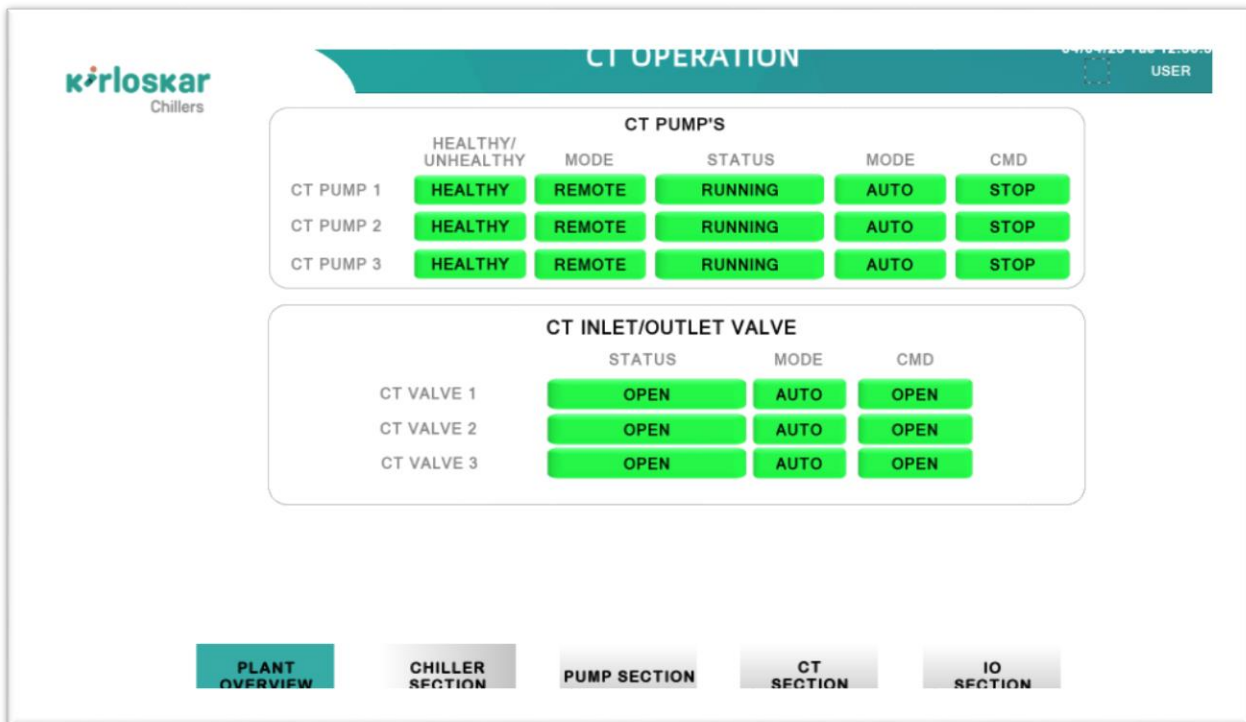
ii. Evaporator Pump 3

If there is a fault while the device is running, then the device will go in fault as shown below and shows the cause by which it is in fault.

6.2.4. Cooling Tower

Cooling Tower screen shows the status of CT fan, CT Inlet valve and CT Outlet Valve of respective sequence.

- i. Cooling Tower 1
- ii. Cooling Tower 2
- iii. Cooling Tower 3



The screenshot displays the 'CT OPERATION' screen with the Korloskar Chillers logo in the top left. The top right corner shows '04/10/2018 14:12:00' and 'USER'. The main content area is divided into two sections: 'CT PUMP'S' and 'CT INLET/OUTLET VALVE'.

CT PUMP'S

	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
CT PUMP 1	HEALTHY	REMOTE	RUNNING	AUTO	STOP
CT PUMP 2	HEALTHY	REMOTE	RUNNING	AUTO	STOP
CT PUMP 3	HEALTHY	REMOTE	RUNNING	AUTO	STOP

CT INLET/OUTLET VALVE

	STATUS	MODE	CMD
CT VALVE 1	OPEN	AUTO	OPEN
CT VALVE 2	OPEN	AUTO	OPEN
CT VALVE 3	OPEN	AUTO	OPEN

At the bottom, there is a navigation bar with five buttons: 'PLANT OVERVIEW' (highlighted in teal), 'CHILLER SECTION', 'PUMP SECTION', 'CT SECTION', and 'IO SECTION'.

6.2.5. Condenser Pump

Condenser screen shows the status of Condenser pumps and Heat pump condenser on respective screens.

- i. Condenser Pump 1
- ii. Condenser Pump 2
- iii. Condenser Pump 3

PUMP OPERATION

EVAPORATOR PUMP'S

	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
EVAP PUMP 1	HEALTHY	REMOTE	RUNNING	AUTO	STOP
EVAP PUMP 2	HEALTHY	REMOTE	RUNNING	AUTO	STOP
EVAP PUMP 3	HEALTHY	REMOTE	RUNNING	AUTO	STOP

CONDENSER PUMP'S

	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
COND PUMP 1	HEALTHY	REMOTE	RUNNING	AUTO	STOP
COND PUMP 2	HEALTHY	REMOTE	RUNNING	AUTO	STOP
COND PUMP 3	HEALTHY	REMOTE	RUNNING	AUTO	STOP

SECONDARY PUMP'S

	HEALTHY/ UNHEALTHY	MODE	STATUS	MODE	CMD
SEC PUMP 1	HEALTHY	REMOTE	RUNNING	AUTO	STOP
SEC PUMP 2	HEALTHY	REMOTE	RUNNING	AUTO	STOP
SEC PUMP 3	HEALTHY	REMOTE	RUNNING	AUTO	STOP

Condenser Pump Overview

PLANT OVERVIEW CHILLER SECTION PUMP SECTION CT SECTION IO SECTION

6.2.6. Chillers

Chiller screen shows the status of Chillers, Evaporator valve, Condenser valve, and Heat pump evaporator on respective screens.

CHILLER AND VALVE OPERATION

04/09/2018 12:30:39 USER

CHILLER OPERATION

	MODE	STATUS	CMD
CHILLER - 1	REMOTE	RUNNING	STOP
CHILLER - 2	REMOTE	RUNNING	STOP
CHILLER - 3	REMOTE	RUNNING	STOP

EVAPORATOR VALVE

	STATUS	MODE	CMD
EVAPORATOR VLAVE 1	OPEN	AUTO	OPEN
EVAPORATOR VLAVE 2	OPEN	AUTO	OPEN
EVAPORATOR VLAVE 3	OPEN	AUTO	OPEN

CONDENSER VALVE

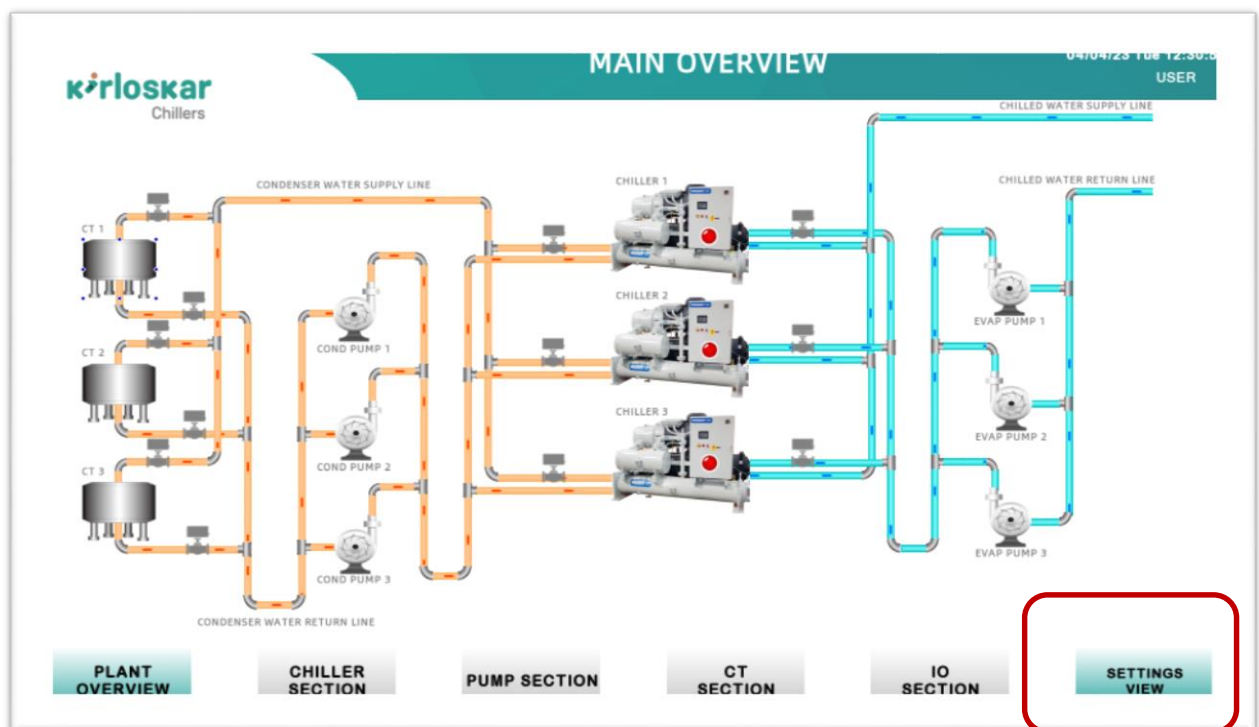
	STATUS	MODE	CMD
CONDENSER VALVE 1	OPEN	AUTO	OPEN
CONDENSER VALVE 2	OPEN	AUTO	OPEN
CONDENSER VALVE 3	OPEN	AUTO	OPEN

Chiller Operation

← PLANT OVERVIEW CHILLER SECTION PUMP SECTION CT SECTION IO SECTION

6.2.7. Settings

Settings screen shows the setting of various parameters for operation of the plant manager in AUTO Mode. Click on Set button to open Settings Page.



After click on setting button open following screen.

Setting
button

CHILLER, VALVE AND PUMP FEEDBACK DELAY							
CHILLER FEEDBACK DELAY				EVAP. VALVE FEEDBACK DELAY			
TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3	TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3
ON F/B DELAY	123.4	123.4	123.4	ON F/B DELAY	123.4	123.4	123.4
OFF F/B DELAY	123.4	123.4	123.4	OFF F/B DELAY	123.4	123.4	123.4
COND. VALVE FEEDBACK DELAY				CT VALVE FEEDBACK DELAY			
TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3	TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3
ON F/B DELAY	123.4	123.4	123.4	ON F/B DELAY	123.4	123.4	123.4
OFF F/B DELAY	123.4	123.4	123.4	OFF F/B DELAY	123.4	123.4	123.4
PUMP ON/OFF FEEDBACK DELAY							
TIMER (Sec)	EVAP. PUMP		COND. PUMP		SECON. PUMP		CT FAN
ON F/B DELAY	123.4		123.4		123.4		123.4
OFF F/B DELAY	123.4		123.4		123.4		123.4

1. Plant Manager Start/Stop Button

Operation of Plant Manager in AUTO Mode is only possible when the START button is pressed; else AUTO operation for plant manager will not be executed. When Plant manager is in Auto and demand is 0 then Evaporator pump 1 and evaporator valve1 are ON for circulation of air.



2. Header Temperature SP and Deadband

When Header temperature is greater than Header Temperature SP + Deadband/2 then the plant manager is in LOADING ZONE and when Header temperature is less than Header Temperature SP - Deadband/2 then the plant manager is in UNLOADING ZONE.

LOADING ZONE – When plant manager is in loading zone then chiller units are turned ON as per the time settings, sequence and availability of the chillers.

Start Sequence / Loading Sequence: -

- **Start**
- **Evaporator Pump**
 - Preset Delay
- **Evaporator Valve**
 - Preset Delay
- **Condenser Pump**
 - Preset Delay
- **Condenser Valve**

- Preset Delay
- **CT valve**
 - Preset Delay
- **Chiller**
 - Preset Delay
- **CT Fan**

UNLOADING ZONE – When plant manager is in unloading zone then chiller units are turned OFF as per the time settings and sequence of the chillers.

Stop Sequence / Unloading Sequence: -

- **Stop**
- **Chiller**
 - Preset Delay
- **CT Valve**
 - Preset Delay
- **Condenser Valve**
 - Preset Delay
- **CT Fan**
 - Preset Delay
- **Condenser Pump**
 - Preset Delay
- **Evaporator Valve**
 - Preset Delay
- **Evaporator Pump**

2. Fast Load Start Button

Fast Load start button is used when the plant manager is in loading zone and the timer for next chiller ON is counting, then on pressing this button the counting will be bypassed and the start sequence for next chiller in sequence will be started.

When 2 chillers are ON and the plant manager goes in Unloading zone then when the % load of the ON chillers goes below this SP then the stop sequence for next Chiller in sequence will start after the count of OFF delay SP is completed.

1. % Load Max

When Plant Manager is in Loading zone and % Load of the ON chiller goes above this SP then the sequence to start the next chiller in sequence is started after the count of ON Delay SP.

2. Next Chi. ON Delay time

If the Plant Manager is in loading zone and ON Chiller % loading does not reach the % Load Max SP for this Specified time, then the sequence to start the next chiller in sequence is started after the count of ON Delay SP. Set Point is in minutes.

3. ON Delay SP

When the ON Chiller % loading is more then % Load SP or Next Chi. ON Delay time is elapsed then the plant manager counts the set minutes and after the time is elapsed, Plant manager starts the ON sequence for next chiller. Set Point is in minutes.

4. OFF Delay SP

When Plant manager goes in Unloading zone then it counts the set minutes and after the time is elapsed, Plant manager starts the stop sequence for next chiller in sequence. Set Point is in minutes.

- a) Set Point 2

1. IN Temp SP and ON Hyst

When the Condenser water in temperature is below the Temp SP then the CT fan will not be started and When the Condenser water in temperature is above Temp SP +ON Hyst then only CT fan will start. Set Point is in degree Celsius.

2. Manual OFF Delay

When all three chillers are running and any one of device which is running in auto mode, is switched to Manual mode then this delay comes in picture. The timers start counting for the set time, before the timer is elapsed all the device of any one sequence should be switched to Manual mode, else it will cause tripping of the chiller. Set Point is in seconds.

i. Chiller Valve Setting

The following screen shows you the Valve feedback time setting for respective Evaporator valve, Condenser valve, CT inlet valve and CT outlet valve. When the output command is close and the feedback for valve close is not received by the system before the time is elapsed then the respective valve will go in fault i.e. FTC (Fail to Close). And when the output command is open and the feedback for valve open is not received by the system before the time is elapsed then the respective valve will go in fault i.e. FTO (Fail to Open). Set Point is in seconds.

CHILLER, VALVE AND PUMP FEEDBACK DELAY							
CHILLER FEEDBACK DELAY				EVAP. VALVE FEEDBACK DELAY			
TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3	TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3
ON F/B DELAY	123.4	123.4	123.4	ON F/B DELAY	123.4	123.4	123.4
OFF F/B DELAY	123.4	123.4	123.4	OFF F/B DELAY	123.4	123.4	123.4
COND. VALVE FEEDBACK DELAY				CT VALVE FEEDBACK DELAY			
TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3	TIMER (Sec)	CHILLER-1	CHILLER-2	CHILLER-3
ON F/B DELAY	123.4	123.4	123.4	ON F/B DELAY	123.4	123.4	123.4
OFF F/B DELAY	123.4	123.4	123.4	OFF F/B DELAY	123.4	123.4	123.4
PUMP ON/OFF FEEDBACK DELAY							
TIMER (Sec)	EVAP. PUMP	COND. PUMP	SECON. PUMP	CT FAN			
ON F/B DELAY	123.4	123.4	123.4	123.4			
OFF F/B DELAY	123.4	123.4	123.4	123.4			

ii. Chiller Pump Setting

The following screen shows you the Pump ON feedback time and OFF feedback time setting for respective Chiller, Evaporator Pump, Condenser Pump, CT Fan and HP Pumps. The timing for Pump ON and OFF is separate and respective for each Pump. When the output command is stop and the feedback for Pump stop is not received by the system before the OFF time is elapsed then the respective Pump will go in fault i.e. FTS (Fail to Stop). And when the output command is start and the feedback for Pump run is not received by the system before the ON time is elapsed then the respective Pump will go in fault i.e. FTR (Fail to Run). Set Point is in seconds.

iii. Delay Setting

The following screen shows you Delay time for start sequence and for stop sequence the delay for respective valve and pump to start and stop are displayed in the screen which can be changed as per requirement. If operator requires that there should be no delay between the two devices to start or stop then the ON or OFF delay for second device should be made to zero. Set Point is in seconds.

iv. Set Date and Time

The following screen allows you set the date and time of the HMI if it lags or leads the actual time.



6.2.8. Alarm

Alarm screen shows the current alarm of the system.

[illegible]

6.2.9. Modbus Data

Modbus data of the chillers will be displayed in the following screen.

